

COAL.

Problem Set 2

Instructions:

This problem set contains 13 questions. Some of these questions are for your practice and some are to be submitted as an assignment. The list of the questions that are to be submitted as assignment is as follow.

Assignment questions: 2, 3, 6, 7, 9, 10, 13

Due date: 17th September, 2026

Submission format: Hand written. You can print out question 3, 9 and 10.

Late Submission: -25% for each day .

Physical Address:

1. Solve Q11 to Q 21 Chapter 1 of Bilal Hashmi
2. Consider the value of segment registers as follow

CS= 0AFFh

DS= FFFFh

ES= F001h

SS=01BBh

Answer the following questions

- a. If bx=0101h, what is the physical address of [bx]?
- b. If bx=0101h what is the physical address of [CS:bx]?
- c. If IP= 0100h, from which physical address the next instruction will be fetched?
- d. If BP= 0FFFh from which physical address is [BP+2]?

3. For each of the code snippet given below, write value of each flag after logical/arithmetic operation and mention if the jump will be taken or not

CODE	SF	OF	CF	ZF	Jump Taken?
MOV AX, 0XF0F0 AND AX, 0X0F0F JZ LABEL1					
MOV AX, 0XF0F0 OR AX, 0XFF0F JS LABEL1					
MOV AX, 0XF2 SUB AX, 0XFA JNS LABEL1					
MOV BX, 0XA2 CMP BX, 0XC0 JL LABEL1					
MOV AX, 5 CMP AX, 6 JL L1					
MOV AX, 5 CMP AX, 4 JG L1					
MOV DX, 0XA523 CMP DX, 0XA523 JE L1					
MOV DX, 0XA523 CMP DX, 0XA523 JNE L5					
MOV AL, +127 ; HEXADECIMAL VALUE IS 7FH CMP AL, -128 ; HEXADECIMAL VALUE IS 80H JG ISGREATER					
MOV AL, +127 ; HEXADECIMAL VALUE IS 7FH CMP AL, -128 ; HEXADECIMAL VALUE IS 80H JA ISABOVE					
MOV AL, +127 ; HEXADECIMAL VALUE IS 7FH CMP AL, -128 ; HEXADECIMAL VALUE IS 80H JNG LABEL					
MOV AL, +127 ; HEXADECIMAL VALUE IS 7FH CMP AL, -128 ; HEXADECIMAL VALUE IS 80H JNA LABEL					
MOV Ax, 0xFFFF SUB AX, 2 ; JG LABEL					
MOV Ax, 0xFFFF SUB AX, 2 ; JA LABEL					

4. What is the highest and the lowest possible signed number that can be stored in 8 bits
5. What is the highest and lowest unsigned number that can be stored in 8 bits
6. What is the highest and the lowest possible signed number that can be stored in 16 bits
7. What is the highest and lowest unsigned number that can be stored in 16 bits

8. Write a program to find square of 'num' and saves it in 'square'. You will find the square of *num* by adding it *num* times. The addition should be performed in a loop.

The basic structure of code is given write you code where specified

```
[org 0x0100]

; your code should start here

; your code ends here

mov ax, 0x4C00

int 21h

num: dw 16

square: dw 0
```

9. Write a program in assembly to find the square of each number in array1 and store it in array 2, using loops. The size of array 1 and array 2 is 10. Note that array 1 is unsigned array and you can only use addition to find the square (use of multiplication instruction is not allowed).

The basic structure of code is given write you code where specified

```
[org 0x0100]

; your code should start here

; your code ends here

mov ax, 0x4C00

Array1: dw 3,5,7,4,9,2,1,0,12,16

Array 2: dw 0,0,0,0,0,0,0,0,0,0
```

10. Write a program in assembly to find the maximum number from a signed array, array 1. Write that number in max. The basic structure of code is given write you code where specified.

```
[org 0x0100]

; your code should start here

; your code ends here

mov ax, 0x4C00

Array1: dw 3,-5,7,4,-9,2,1,0,12,-16

Max: dw 0
```

11. How will you modify above code if the array was unsigned?

12. How will you modify the following code using unconditional jump such that it works properly?

```
num1: dw 1,  
num2: db 10  
sum: dw 0  
  
[org 0x0100]  
    mov ax, [num1]  
    mov bl, [num2]  
    add ax, bx  
    mov [sum], ax  
  
    mov ax, 0x4c00 ; terminate program  
    int 0x21
```

13. Translate the following flow charts to assembly code

